Yoruba Drum Notation Language (YDNL): A Software Framework for Encoding and Preserving Yorùbá Polyrhythmic Music

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Abstract

Yorùbá drum traditions encode language, ritual, and genealogical memory, yet no existing software standard can represent this music in machine-readable form. The dominant symbolic encoding standards, MusicXML, the Music Encoding Initiative (MEI) format, and Humdrum, assume Western staff notation and cannot carry independently layered metric cycles, continuously gliding pitch used as a speech surrogate, stroke timbre as semantic content, or the relational structure of a drum ensemble. This paper introduces the YorubaDrum Notation Language (YDNL), an open, formally specified data schema, file format, and web prototype for Yorùbá Dùndún and Bàtá drum music, developed through Design Science Research. YDNL models a pattern as concurrent rhythmic layers with independent cycles, tonal registers with optional glides, a controlled stroke-timbre vocabulary, relational ensemble roles, and mandatory cultural metadata. The framework was evaluated over a twelve-pattern synthetic reference corpus using metrics fixed before development. YDNL achieved a mean Encoding Completeness Score of 1.00 against a seven-feature structural taxonomy, whereas a fairness-preserving MusicXML benchmark encoding of the same patterns retained a mean of 0.33, losing on average 4.2 of 7 structural features per pattern. The Benchmark Parity Ratio averaged 2.08, meaning YDNL preserves about twice as many structural features as it shares with MusicXML, and faceted retrieval precision was 1.00. The central finding is that the encoding gap is architectural rather than incidental: closing it requires a schema designed outward from

the music. A fieldwork corpus of canonical patterns and practitioner validation with master drummers are in progress.

Keywords: Yorùbá drum music, polyrhythmic encoding, digital cultural heritage, music notation language, Design Science Research, MusicXML

1. Introduction

The Dùndún talking drum is among the most sophisticated communication technologies of the Yorùbá people of south-west Nigeria. By squeezing the leather cords of its hourglass body, the player reproduces the three tone levels of the Yorùbá language and the glides between them, with enough fidelity that acoustic analysis confirms a strong correspondence between drummed and spoken Yorùbá (Durojaye, Knowles, Patten, Garcia, & McBeath, 2021). The Bàtá ensemble, led by the ìyá ìlù, has been the ritual instrument of Sango worship for centuries (Metropolitan Museum of Art, 2024), mapping open, muted, and slap strokes onto low, mid, and high speech tones so that timbre itself carries linguistic meaning (González & Oludare, 2021; Villepastour, 2010). In both traditions the rhythms encode genealogies, proverbs, and theology inseparable from the patterns that carry them (Euba, 1990).

This knowledge lives almost entirely in human memory. It is transmitted through the Àyàn lineage with no written or machine-readable form, and the decline of trained specialists places it at measurable risk (British Museum Endangered Material Knowledge Programme, 2025; Tanure & Nzinga, 2023; UNESCO, 2003). The scholarly record is deep (Agawu, 2016; Arom, 1991; Euba, 1990; Villepastour, 2010), but the software infrastructure to store, validate, exchange, and compute on these patterns does not exist. The difficulty is one of representation. MusicXML targets common Western notation, and its own documentation states that non-Western traditions would be better served by a dedicated language (Good, 2001; W3C Music Notation Community Group, 2021); MEI is extensible, but its established profiles assume the European stave (Music Encoding Initiative, 2023); and the Humdrum kern format encodes the core information of common-practice Western music (Huron, 1994). None can represent independently layered metric cycles, gliding pitch used as a linguistic surrogate, timbre as meaning, or ensemble interdependence.

This paper addresses that gap. Its objective is fourfold: to analyse the encoding gap precisely, to design a data model, schema, and file format that closes it, to build a working web prototype that implements the schema, and to evaluate the result against metrics fixed before development. The work follows Design Science Research (Hevner, March, Park, & Ram, 2004) and is organised around a Drum Pattern Corpus, so that the schema is shaped by the music rather than the music forced into the schema. The study contributes five citable artefacts: an open Drum Pattern Corpus programme with a published twelve-pattern reference corpus; a Structural Incompatibility Report; the YDNL specification (data model, XML Schema, and JSON Schema); an open-source web prototype that encodes, validates, renders, plays, retrieves, and computationally transcribes patterns; and an evaluation framework of five metrics with thresholds fixed in advance, three of which are already measured and reported here. To the best of the authors' knowledge, YDNL is the first open, machine-readable encoding framework built specifically for Yorùbá drum music.

The remainder of this paper is organised as follows. Section 2 reviews related work and states the research gap. Section 3 presents the methodology, corpus programme, and evaluation framework. Section 4 presents the YDNL schema, the prototype, and the measured results. Section 5 discusses the findings, and Section 6 concludes and states the limitations and future work.

2. Literature Review

2.1 Symbolic Music Encoding Standards

Three symbolic formats dominate digital music, and all assume Western notation. MusicXML, the principal interchange format for scores, was designed for common Western music notation from the seventeenth century onwards, and its designers explicitly note that other notational traditions would be better served by dedicated languages (Good, 2001); the current MusicXML 4.0 specification retains this scope (W3C Music Notation Community Group, 2021). MEI is extensible through profiling, but its established profiles describe stages of European notation history such as mensural and neume notation (Music Encoding Initiative, 2023). The Humdrum kern representation captures the core information of common-practice Western music (Huron, 1994). Each format is therefore bound to one notational world view, and the assumptions that bind them, a single metric hierarchy, discrete chromatic pitch, and timbre as an expressive annotation rather than a semantic unit, are precisely the assumptions Yorùbá drum music violates.

2.2 African Rhythm and Yorùbá Drum Scholarship

The musicological record of the Yorùbá drum traditions is rich. Euba (1990) documented the Dùndún tradition comprehensively, and Villepastour (2010) decoded the Bàtá ensemble's stroke vocabulary as a systematic code. The acoustic and phonological basis of both drums as speech surrogates is empirically established: Durojaye, Knowles, et al. (2021) measured the correspondence between drummed and spoken Yorùbá; Durojaye, Fink, Roeske, Wald-Fuhrmann, and Larrouy-Maestri (2021) showed that listeners perceive dùndún performance on a continuum between speech-like and music-like modes; and González and Oludare (2021) analysed the interface between the organology of the two drum families and Yorùbá phonology. Comparable African speech surrogates such as the Sambla balafon have received similar grammatical analysis (McPherson, 2018). On the rhythmic side, Arom (1991) provided the canonical cyclic analysis of African polyrhythm, and Agawu (2016) argued that African rhythmic practice constitutes a coherent theoretical system in its own right rather than a deviation from Western metre. Frishkopf (2021) tested staff notation, tabular notation, and a recursive grammar against the Ewe Agbekor tradition, found each inadequate, and called explicitly for notation innovation. This body of work documents the knowledge richly but offers no machine-readable schema or open file format.

2.3 Computational Ethnomusicology and Digital Heritage

Music information retrieval was built mainly around Western music, and researchers have long noted that culture-specific representations are required for other traditions (Tzanetakis, Kapur, Schloss, & Wright, 2007). Surveys of world-music corpus research confirm that the availability of suitable symbolic representations, not the availability of algorithms, is the persistent bottleneck (Panteli, Benetos, & Dixon, 2018). Even where automatic transcription tools exist, user studies with ethnomusicologists show that fully automatic output is not trusted for scholarly use and that human-in-the-loop workflows are required (Holzapfel & Benetos, 2019). In the heritage domain, recent reviews of intangible cultural heritage digitisation identify structured, machine-actionable representation of embodied knowledge as the open problem, beyond simple audiovisual archiving (Hou, Kenderdine, Picca, Egloff, & Adamou, 2022). These findings jointly motivate the present work's ordering: an explicit, validatable representation must precede any computational or machine-learning application.

2.4 The Research Gap

Existing systems thus split into rigorous standards that cannot represent the music and rich documentation that is not machine-readable. The closest system, AfriScore, is a culturally informed web-based learning platform for scoring African music nuances, but it publishes no formal schema or open standard ("Introducing AfriScore," 2025). Table 1 maps the requirements of Yorùbá drum music against the available options: no existing standard meets more than two of the seven requirements, and those two concern file-format openness rather than the music itself. YDNL targets the unoccupied intersection: a formally defined, open, machine-readable framework grounded in Yorùbá musicology and designed for validation by Nigerian practitioners.

Table 1

Capability Comparison of Existing Encoding Standards Against YDNL

Encoding requirement	MusicXML	MEI	AfriScore	YDNL
Polyrhythm and independent metric layering	No	No	Partial	Yes
Continuously variable pitch / speech surrogacy	No	No	No	Yes
Stroke timbre as semantic content	No	No	No	Yes
Ensemble relational interdependence model	No	No	No	Yes
Ritual and cultural metadata as first-class data	No	No	No	Yes
Open machine-readable file format	Yes	Yes	No	Yes
Designed for African musical traditions	No	No	Partial	Yes

3. Methodology

3.1 Design Science Research Framing

The framework is built using Design Science Research (DSR), the paradigm for research whose primary output is an artefact (Hevner et al., 2004). Figure 1 places the work within the three DSR cycles: the Relevance cycle ties it to the living practice of master drummers and the cultural context of each pattern; the Rigor cycle grounds it in ethnomusicology, encoding standards, and software engineering; and the Design cycle builds and evaluates the artefacts, which span all four DSR artefact types (construct, model, method, and instantiation).

[Figure 1 — paste diagram image here]

Figure 1. The Work Positioned Within the Three Design Science Research Cycles, With the Drum Pattern Corpus as the Shared Artefact of Record

3.2 The Drum Pattern Corpus and Data Integrity

A Drum Pattern Corpus forms the backbone of the programme. The target collection comprises twenty-four canonical patterns, twelve Dùndún and twelve Bàtá, selected in collaboration with the Department of Music and Creative Arts at the Federal University of Technology, Akure, and master drummers in Ondo and Oyo States, to cover at least six Orisha associations, the principal metric ratios 3:4, 3:8, and 4:12, the documented stroke timbres, and ensembles of two to five drums. Each fieldwork pattern is captured in three independent forms before encoding, an audio field recording, a faculty notation sketch, and a structured textual description, giving every encoding a ground truth independent of the schema.

Because the fieldwork collection is still in progress, the schema and prototype are exercised in this paper over a published reference corpus of twelve synthetic patterns, six Dùndún and six Bàtá, constructed to exercise every feature of the schema. Data integrity is enforced structurally: every synthetic pattern carries an empty recording reference, an empty drummer attribution, and the validation status draft, so the provenance state of every record is visible in the data itself. Provenance fields are populated, and validation status advances, only when a real recording, a real attribution, and a real review event exist. This makes the distinction between illustrative and field-validated data machine-checkable rather than a matter of documentation alone.

3.3 Research Phases

The work proceeds in five phases. Phase 0 builds and openly publishes the corpus, so that it functions as the shared artefact of record. Phase 1 conducts the domain analysis and produces a Structural Incompatibility Report documenting, pattern by pattern, which features MusicXML and MEI cannot encode and why. Phase 2 designs the data model, the XML and JSON schemas, and the controlled cultural vocabulary, and fixes every evaluation metric and threshold in advance, on the principle that metrics defined after results are rationalisations rather than measures. Phase 3 builds the web prototype. Phase 4 computes the metrics and conducts validation sessions with master drummers, music educators, and an external ethnomusicologist. No machine learning is used in any phase, by design: the contribution is an explicit and inspectable representation that practitioners can validate, and the corpus is a precondition for any future learning-based work rather than a product of it. Phases 0 through 3 and the structural evaluation of Phase 4 are complete for the reference corpus; the fieldwork collection and practitioner sessions are in progress.

3.4 Evaluation Framework

Five metrics were fixed before the prototype was built (Table 2). The primary structural measure is the Encoding Completeness Score (ECS), ECS = |Fenc| / |Fpresent| (1), the fraction of the structural features present in a pattern that a given encoding can carry, computed against a seven-feature taxonomy derived from the domain analysis (Section 4.1). For the fieldwork corpus, feature presence is scored by two coders, one from software engineering and one from ethnomusicology, once inter-coder reliability reaches a Cohen kappa of 0.80; for the reference corpus, feature presence is detected deterministically from the encoded data. The Benchmark Parity Ratio (BPR), BPR = |FYDNL-unique| / |Fshared| (2), divides the number of a pattern's structural features that only YDNL can carry by the number that YDNL shares with the benchmark standard; a value above 2.0 means that for every structural feature the Western encoding retains, YDNL preserves more than two additional features that it loses. When a pattern shares no features with the benchmark, the denominator is clamped to one, a convention that understates rather than overstates the disparity. The remaining three metrics place authority outside the software: the Practitioner Accuracy Rating with master drummers, the System Usability Scale with music educators, and Retrieval Precision over faceted queries on the corpus.

Table 2

YDNL Evaluation Metrics, Fixed in Advance of Prototype Development

Metric	Definition	Target	Status
Encoding Completeness Score (ECS)	Features carried by an encoding over features present; dual-coded (Cohen kappa ≥ 0.80) for fieldwork patterns.	ECS ≥ 0.85	Measured (Sec. 4.5)
Benchmark Parity Ratio (BPR)	YDNL-unique features captured over features shared with the benchmark standard.	BPR > 2.0	Measured (Sec. 4.5)
Practitioner Accuracy Rating (PAR)	Master drummers rate rhythmic, tonal, timbral, and cultural accuracy (5-point scale).	Mean ≥ 4.0	In progress
System Usability Scale (SUS)	Standard 10-item SUS with music educators.	SUS ≥ 68	In progress
Retrieval Precision (RP)	Precision of library retrieval by Orisha, ceremony, instrument, or region.	RP ≥ 0.90	Measured (Sec. 4.5)

4. Results

4.1 The Structural Feature Taxonomy

The Phase 1 domain analysis distilled the requirements of Dùndún and Bàtá music into a taxonomy of seven structural features, which then served as the fixed measurement frame for every metric: (a) multiple concurrent rhythmic layers; (b) polymeter, that is, layers with differing metric cycle lengths; (c) discrete low, mid, and high tonal registers; (d) continuous pitch glides used as a speech surrogate; (e) stroke timbre variation carrying semantic content; (f) a relational ensemble structure of interdependent roles; and (g) ritual and cultural metadata as structural content. MusicXML can carry two of the seven, concurrent layers and discrete registers, even if only in degraded form; the Structural Incompatibility Report documents why the remaining five have no representation in MusicXML or MEI.

4.2 Schema Design Principles

Five principles shape the YDNL schema, each answering one part of the taxonomy. Metric plurality gives each rhythmic layer its own cycle length, pulse unit, and offset against a shared reference pulse, so no pattern-level barline is imposed: a pattern is modelled as a set of concurrent layers rather than one metric stream. Tonal continuity models pitch as a register (low, mid, high) with an optional glide carrying start and end registers, direction, and duration, and no chromatic pitch class, in keeping with the dùndún's linguistic function (Durojaye, Knowles, et al., 2021). Timbral semantics makes stroke type a mandatory attribute drawn from a controlled vocabulary, open, muted, slap, heel-toe, rim, and combined, each mapped to a documented technique (Villepastour, 2010). Ensemble relationality models drums as named role nodes with interdependency rules. Cultural integrity makes a cultural annotation block mandatory, so a file lacking ceremonial and provenance metadata fails validation outright: such a file is treated as incomplete data, not as complete data with optional metadata.

4.3 Data Model and Serialisation

Table 3 lists the core entities of the model. The DrumPattern is the root aggregate and the unit of exchange; it owns one or more RhythmicLayers, each owning an ordered sequence of StrokeEvents that may carry a PitchGlide; the EnsembleConfig is referenced, and the CulturalAnnotation is embedded. YDNL serialises primarily to XML with a parallel JSON schema, and the formal XSD and JSON Schema are published openly so validation runs in any standard toolchain.

Table 3

Core Entities of the YDNL Data Model

Entity	Role in model	Core attributes
DrumPattern	Root aggregate, unit of exchange	pattern_id, title, tempo_bpm, metric_ratio, ensemble_ref, cultural_annotation
RhythmicLayer	Child of DrumPattern	layer_id, instrument_role, pulse_unit, metric_cycle_length, offset_pulses
StrokeEvent	Child of RhythmicLayer	event_id, pulse_position, stroke_type, pitch_register, duration_pulses
PitchGlide	Attribute type on StrokeEvent	start_register, end_register, glide_direction, duration_pulses
EnsembleConfig	Referenced entity	ensemble_id, instrument_nodes, interdependency_rules
CulturalAnnotation	Embedded in DrumPattern	orisha_id, ceremony_type, regional_variant, source_drummer_id, validation_status

The validator also enforces organological constraints from the domain analysis. Only the hourglass tension drums of the Dùndún family (ìyá ìlù dùndún, gangan, kerikeri) can bend pitch, because the player squeezes the lacing cords; Bàtá drums are fixed-pitch, tuned with ida wax paste, and render speech through stroke timbre instead, and the gudugudu is a fixed-pitch kettle drum (González & Oludare, 2021). A pitch glide encoded on any fixed-pitch role is therefore rejected as a validation error, so the schema cannot silently record a performance that the instrument cannot physically produce.

Listing 1 encodes a partial Dùndún festival pattern from the reference corpus and exercises the five principles at once: the cultural annotation is embedded at the root, the two layers run independent twelve- and eight-pulse cycles, and the event at pulse nine carries a pitch glide that no existing standard can express. Consistent with Section 3.2, the drummer attribution is empty and the validation status is draft, because this synthetic pattern has not been performed or reviewed by any drummer.

```xml
<?xml version="1.0" encoding="UTF-8"?>
<ydnl version="1.0" xmlns="https://ydnl.org/schema/v1">
  <drumPattern id="DP-0021" title="Dundun festival pattern">
    <culturalAnnotation>
      <orisha>Sango</orisha> <ceremony_type>festival</ceremony_type>
      <region>Oyo</region> <source_drummer></source_drummer>
      <validation_status>draft</validation_status>
    </culturalAnnotation>
    <ensembleRef id="ENS-Dundun-2" />
    <rhythmicLayers>
      <layer instrument_role="iya-ilu-dundun"
             metric_cycle_length="12" pulse_unit="semiquaver">
        <strokeEvent pulse="1" stroke_type="open" pitch_register="high" />
        <strokeEvent pulse="5" stroke_type="muted" pitch_register="low" />
        <strokeEvent pulse="9" stroke_type="open" pitch_register="glide">
          <pitchGlide start="mid" end="high"
                      direction="ascending" duration_pulses="2" />
        </strokeEvent>
      </layer>
      <layer instrument_role="gudugudu"
             metric_cycle_length="8" pulse_unit="semiquaver">
        <strokeEvent pulse="1" stroke_type="open" pitch_register="low" />
        <strokeEvent pulse="4" stroke_type="rim" pitch_register="mid" />
      </layer>
    </rhythmicLayers>
  </drumPattern>
</ydnl>
```

Listing 1. A Representative YDNL XML Excerpt for a Dùndún Festival Pattern. The two layers run independent 12- and 8-pulse cycles; the glide at pulse 9 is legal only on a tension drum; provenance fields are empty and the status is draft because the pattern is synthetic.

To make the comparison concrete, consider what happens when the pattern in Listing 1 is exported to MusicXML. Both layers must be placed on a single shared metre, so the independent twelve- and eight-pulse cycles collapse onto one barline and the defining cross-rhythm is lost. The gliding tone on pulse nine has no representation, since MusicXML and MEI quantise pitch to chromatic steps. The open, muted, and rim distinctions, which carry the linguistic content of the drum, are flattened into undifferentiated noteheads. And the Orisha, ceremony, drummer, and validation metadata have nowhere to live in a score model at all. YDNL preserves all four, which is precisely the difference the Benchmark Parity Ratio quantifies.

4.4 System Architecture and Prototype

The prototype uses a layered architecture (Figure 2) governed by a single validation contract. A React interface communicates only over JSON with an Express API whose routes map directly to schema operations; neither contains structural logic, and both delegate every operation on a pattern to a framework-independent core library, ydnl-core, which holds the data model, the validator, the XML and JSON serialisers, a MusicXML benchmark encoder, and the metric functions. Persistence uses a JSON corpus store for the prototype and a PostgreSQL schema mirroring the entity model for production, behind one repository interface. The XSD and JSON Schema act as the single source of truth, enforced at the API boundary and inside the core, keeping the file format, the database, and the runtime checks consistent with one another.

[Figure 2 — paste diagram image here]

Figure 2. Layered Architecture of the YDNL Prototype, With the XSD and JSON Schema Forming a Single Validation Contract Enforced Across the Stack

The prototype implements six components against this core. A graphical timeline encoder places stroke events across up to five concurrent layers and writes the model directly. A file manager imports and exports schema-valid XML and JSON with element-level error reporting. A renderer draws each layer as a row of stroke symbols, shows pitch glides as gradient arrows, and plays the pattern audibly through a sample-accurate lookahead scheduler that advances each layer on its own metric cycle, so the polymeter is heard, not merely drawn. A searchable library gives faceted access by Orisha, ceremony, instrument, metric ratio, and region. A benchmark encoder generates a parallel MusicXML version of any pattern and lists the features the Western format cannot carry, making the comparative claim of this paper demonstrable inside the tool itself. Finally, an assisted-transcription component analyses a recorded audio clip using onset detection over a signal envelope, autocorrelation pitch estimation, glide classification, and decay-based stroke-timbre classification, and emits a draft YDNL pattern for human correction; its rationale is discussed in Section 5.

4.5 Measured Structural Results

Table 4 reports the three structural metrics computed over the twelve-pattern reference corpus; the computation script is published with the source code and reproduces every value in the table. The benchmark encoder is deliberately fair to MusicXML: onset timing, durations, and note values are representable in MusicXML and are preserved in the export, so the losses counted by the metrics are only those the format genuinely cannot express.

Table 4

Structural Evaluation Results Over the Twelve-Pattern Reference Corpus

Measure	Mean value	Target	Outcome
YDNL Encoding Completeness Score	1.00	≥ 0.85	Met
MusicXML Encoding Completeness Score (same patterns)	0.33	—	Contrast measure
Structural features lost per pattern in MusicXML	4.2 of 7	—	Contrast measure
Benchmark Parity Ratio	2.08	> 2.0	Met
Retrieval Precision (four faceted queries)	1.00	≥ 0.90	Met

The per-pattern results show a musically meaningful structure. The six Dùndún patterns that carry pitch glides score a BPR of 2.50: five YDNL-unique features against the two features MusicXML shares. The Bàtá patterns score 2.00 when polymetric and 1.50 otherwise, because bàtá drums are fixed-pitch and correctly carry no glides, so one fewer YDNL-unique feature is present. Every pattern in the corpus loses its timbre semantics, its ensemble relationality, and its cultural metadata in MusicXML; the polymetric patterns additionally lose their independent cycles, and the Dùndún patterns their glides. Retrieval precision was computed over four faceted queries (Orisha, ceremony, instrument, and region), each returning only relevant patterns. The Practitioner Accuracy Rating and the System Usability Scale require the fieldwork corpus and human participants, and are therefore reported as in progress rather than estimated.

5. Discussion

The five metrics divide into two kinds, and reading them correctly matters. The YDNL Encoding Completeness Score of 1.00 and the retrieval precision of 1.00 are verification rather than discovery, since YDNL is defined to cover the whole feature taxonomy and faceted search runs over a controlled vocabulary, so these values confirm only that the implementation does what the specification says. The substantive findings are the contrast measures. A mean MusicXML Encoding Completeness Score of 0.33 means that two-thirds of the structural content of these patterns simply vanishes in the dominant interchange format, and a Benchmark Parity Ratio of 2.08 means that for every structural feature MusicXML retains, more than two are lost. Because the Benchmark Parity Ratio denominator is clamped when no features are shared, the reported ratio understates rather than overstates the disparity.

This pattern of losses supports the paper's central claim that the encoding gap is architectural rather than incidental, because each lost feature fails for a distinct structural reason. Polymeter fails because MusicXML's measure model forces one time signature across the score; pitch glides fail because pitch is quantised to chromatic steps, whereas the dùndún's meaning lives in the continuous trajectory between registers; stroke timbre fails because notation formats treat timbre as an expressive annotation rather than the semantic unit it is in Bàtá practice; and ensemble relationality and cultural metadata fail because a score model has no slot for either. No amount of annotation layered onto a Western score recovers these, and recovering them requires the different primitives that YDNL provides. This corroborates, in measurable form, Frishkopf's (2021) conclusion that existing representations are inadequate for West African polyrhythm and that notational innovation is required.

Beyond representational capability, the framework deliberately locates cultural authority outside the software. The mandatory cultural annotation and the three-valued validation status encode the review lifecycle in the data, and the reference corpus published with this paper carries empty provenance and draft status throughout, because populating attribution or validation fields with values that correspond to no real recording or review event would constitute data fabrication. The organological validation rule extends the same discipline to musical content by refusing encodings that are physically impossible on the instrument. The abstention from machine learning in the core is itself a methodological finding, since learning-based tools for non-Western music fail first at the representation layer; YDNL supplies the missing substrate so that future transcription, retrieval, and generative work has both training data and an evaluable output space, consistent with evidence that ethnomusicologists do not accept fully automatic transcription for scholarly use (Holzapfel & Benetos, 2019).

Finally, because the vocabulary is controlled and the ensemble model is modular, the same framework extends, with new vocabularies and ensemble roles, to other traditions such as Igbo Ogene, Ewe Agbekor, and Hausa Kalangu. Framed against the conference subtheme of digital solutions for national infrastructure, the appropriate unit of adoption is institutional, since an open standard endures only with stewardship by heritage institutions, university music departments, and software engineering groups, which is why the artefacts are published openly rather than held as a closed product.

6. Conclusion, Limitations and Future Work

Yorùbá drum traditions are sophisticated knowledge systems that have been documented extensively but never given the machine-readable infrastructure that Western art music has enjoyed for decades. This paper introduced the YorubaDrum Notation Language, an open, formally specified schema, file format, and working prototype that takes the music as its design domain: independent metric cycles, gliding pitch as a linguistic register, stroke timbre as a semantic attribute, relational ensemble structure, and mandatory cultural metadata, developed through Design Science Research with every evaluation threshold fixed in advance. The measured results meet the two pre-registered structural targets, with a mean Encoding Completeness Score of 1.00 against 0.33 for MusicXML over the same patterns and a Benchmark Parity Ratio of 2.08, and they locate the failure of existing standards at the level of architecture rather than notational detail.

The work is not without limitations. The measured results are computed over a synthetic reference corpus whose feature content is known by construction, which is the correct instrument for measuring representational capability but says nothing yet about how faithfully YDNL captures specific canonical performances. The implemented benchmark encoder targets MusicXML only, while the incompatibility analysis for MEI remains analytic rather than executable. The structural metrics count features as present or lost and do not grade partial degradation, so they understate, for example, the difference between losing polymeter entirely and approximating it with tuplets. Finally, the usability and accuracy metrics that involve human participants are not yet collected, so no claim is made here about practitioner acceptance.

We are addressing these limitations directly in the next stage of the programme. We are looking at enhancing the work in the future by completing the twenty-four-pattern fieldwork corpus and conducting the Practitioner Accuracy Rating and System Usability Scale studies with master drummers and music educators, so that fidelity to canonical performances can be measured alongside representational capability. We further intend to implement an executable MEI benchmark, extend the corpus to additional Yorùbá sub-traditions and diaspora Bàtá variants, build profiles for other West African drum families such as Igbo Ogene, Ewe Agbekor, and Hausa Kalangu on the same foundation, and apply machine learning to the resulting dataset, which the present infrastructure makes possible rather than presupposes. Beyond our own programme, we recommend that heritage institutions, university music departments, and software engineering groups adopt and extend open standards of this kind as shared preservation infrastructure, since an open standard endures only through sustained institutional stewardship.

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